

Definite Integral

6.1 Definition

Let $\phi(x)$ be the primitive or anti-derivative of a function $f(x)$ defined on $[a, b]$ i.e., $\frac{d}{dx}[\phi(x)] = f(x)$. Then the definite integral of $f(x)$ over $[a, b]$ is denoted by $\int_a^b f(x)dx$ and is defined as $[\phi(b) - \phi(a)]$ i.e., $\int_a^b f(x)dx = \phi(b) - \phi(a)$. This is also called Newton Leibnitz formula.

The numbers a and b are called the limits of integration, ' a ' is called the lower limit and ' b ' the upper limit. The interval $[a, b]$ is called the interval of integration. The interval $[a, b]$ is also known as range of integration.

Important Tips

☞ In the above definition it does not matter which anti-derivative is used to evaluate the definite integral, because if $\int f(x)dx = \phi(x) + c$, then

$$\int_a^b f(x)dx = [\phi(x) + c]_a^b = (\phi(b) + c) - (\phi(a) + c) = \phi(b) - \phi(a).$$

In other words, to evaluate the definite integral there is no need to keep the constant of integration.

☞ Every definite integral has a unique value.

Example: 1

$$\int_{-1}^3 \left[\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right] dx =$$

- (a) π (b) 2π (c) 3π (d) None of these

[Karnataka CET 2000]

Solution: (b)

$$I = \int_{-1}^3 \left[\tan^{-1} \frac{x}{x^2+1} + \cot^{-1} \frac{x}{x^2+1} \right] dx$$

$$\Rightarrow I = \int_{-1}^3 \frac{\pi}{2} dx \Rightarrow I = \frac{\pi}{2} [x]_{-1}^3 = \frac{\pi}{2} [3+1] = 2\pi.$$

Example: 2

$$\int_0^{\pi} \sin^2 x dx \text{ is equal to}$$

- (a) π (b) $\pi/2$ (c) 0 (d) None of these

[MP PET 1999]

Solution: (b)

$$I = \frac{1}{2} \int_0^{\pi} 2 \sin^2 x dx = \frac{1}{2} \int_0^{\pi} [1 - \cos 2x] dx$$

$$\Rightarrow I = \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi} \Rightarrow I = \frac{1}{2} [\pi] = \frac{\pi}{2}.$$

6.2 Definite Integral as the Limit of a Sum

Let $f(x)$ be a single valued continuous function defined in the interval $a \leq x \leq b$, where a and b are both finite. Let this interval be divided into n equal sub-intervals, each of width h by inserting $(n - 1)$ points $a + h, a + 2h, a + 3h, \dots, a + (n - 1)h$ between a and b . Then $nh = b - a$.

$$\begin{aligned}
 \text{Now, we form the sum } & hf(a) + hf(a+h) + hf(a+2h) + \dots + hf(a+rh) + \dots + hf[a+(n-1)h] \\
 &= h[f(a) + f(a+h) + f(a+2h) + \dots + f(a+rh) + \dots + f(a+(n-1)h)] \\
 &= h \sum_{r=0}^{n-1} f(a+rh)
 \end{aligned}$$

where, $a+nh = b \Rightarrow nh = b-a$

The $\lim_{h \rightarrow 0} h \sum_{r=0}^{n-1} f(a+rh)$, if it exists is called the **definite integral** of $f(x)$ with respect to x between the limits a and b

and we denote it by the symbol $\int_a^b f(x) dx$.

$$\text{Thus, } \int_a^b f(x) dx = \lim_{h \rightarrow 0} h[f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h)] \Rightarrow \int_a^b f(x) dx = \lim_{h \rightarrow 0} h \sum_{r=0}^{n-1} f(a+rh)$$

where, $nh = b-a$, a and b being the limits of integration.

The process of evaluating a definite integral by using the above definition is called integration from the first principle or integration as the limit of a sum.

Important Tips

In finding the above sum, we have taken the left end points of the subintervals. We can take the right end points of the sub-intervals throughout.

$$\text{Then we have, } \int_a^b f(x) dx = \lim_{h \rightarrow 0} h[f(a+h) + f(a+2h) + \dots + f(a+nh)], \Rightarrow \int_a^b f(x) dx = h \sum_{r=1}^n f(a+rh)$$

where, $nh = b-a$.

$$\int_a^b \frac{dx}{\sqrt{(x-\alpha)(\beta-x)}} \quad (\beta > \alpha) = \pi$$

$$\int_a^b \sqrt{(x-\alpha)(\beta-x)} \, dx = \frac{\pi}{8} (\beta-\alpha)^2$$

$$\int_a^b \sqrt{\frac{x-a}{b-x}} \, dx = \frac{\pi}{2} (b-a)$$

$$\int_a^b f(x) \, dx = \frac{1}{n} \int_{na}^{nb} f(x) \, dx$$

$$\int_{a-c}^{b-c} f(x+c) \, dx = \int_a^b f(x) \, dx \quad \text{or} \quad \int_{a+c}^{b+c} f(x-c) \, dx = \int_a^b f(x) \, dx$$

Some useful results for evaluation of definite integrals as limit for sums

$$\text{(i) } 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\text{(ii) } 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\text{(iii) } 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

$$\text{(iv) } a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}, r \neq 1, r > 1$$

$$\text{(v) } a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1 - r^n)}{1 - r}, r \neq 1, r < 1$$

$$\text{(vi) } \sin a + \sin(a+h) + \dots + \sin[a+(n-1)h] = \sum_{r=0}^{n-1} [\sin(a+nh)] = \frac{\sin\left\{a + \left(\frac{n-1}{2}\right)h\right\} \sin\left\{\frac{nh}{2}\right\}}{\sin\left(\frac{h}{2}\right)}$$

$$(vii) \cos a + \cos(a+h) + \cos(a+2h) + \dots + \cos[a+(n-1)h] = \sum_{r=0}^{n-1} [\cos(a+nh)] = \frac{\cos\left\{a + \left(\frac{n-1}{2}\right)h\right\} \sin\left\{\frac{nh}{2}\right\}}{\sin\left(\frac{h}{2}\right)}$$

$$(viii) 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \frac{1}{6^2} + \dots \infty = \frac{\pi^2}{12} \quad (ix) 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \dots \infty = \frac{\pi^2}{6}$$

$$(x) 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \infty = \frac{\pi^2}{8} \quad (xi) \frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \dots \infty = \frac{\pi^2}{24}$$

$$(xii) \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \text{ and } \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad (xiii) \cosh \theta = \frac{e^{\theta} + e^{-\theta}}{2} \text{ and } \sinh \theta = \frac{e^{\theta} - e^{-\theta}}{2}$$

6.3 Evaluation of Definite Integral by Substitution

When the variable in a definite integral is changed, the substitutions in terms of new variable should be effected at three places.

(i) In the integrand

(ii) In the differential say, dx

(iii) In the limits

For example, if we put $\phi(x) = t$ in the integral $\int_a^b f\{\phi(x)\} \phi'(x) dx$, then $\int_a^b f\{\phi(x)\} \phi'(x) dx = \int_{\phi(a)}^{\phi(b)} f(t) dt$.

Important Tips

$$\int_0^{\pi} \frac{dx}{1 + \sin x} = 2$$

$$\int_0^{\pi/2} \log(\tan x) dx = 0$$

$$\int_0^a \frac{dx}{\sqrt{a^2 - x^2}} = \frac{\pi}{2}$$

$$\int_0^{\pi/2} \frac{dx}{\sin x + \cos x} = \sqrt{2} \log(\sqrt{2} + 1)$$

$$\int_0^a \frac{dx}{1 + e^{f(x)}} = \frac{a}{2}, \text{ where } f(a-x) = -f(x)$$

$$\int_0^a \frac{dx}{x^2 + a^2} = \frac{\pi}{2a}$$

$$\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$$

Example: 3 If $h(a) = h(b)$, then $\int_a^b [f(g(h(x)))]^{-1} f(g(h(x))) g'(h(x)) h'(x) dx$ is equal to

[MP PET 2001]

(a) 0

(b) $f(a) - f(b)$

(c) $f[g(a)] - f[g(b)]$

(d) None of these

Solution: (a) Put $f(g(h(x))) = t \Rightarrow f(g(h(x))) g'(h(x)) h'(x) dx = dt$

$$\therefore \int_{f(g(h(a)))}^{f(g(h(b)))} t^{-1} dt = [\log t]_{f(g(h(a)))}^{f(g(h(b)))} = 0 \quad [\because h(a) = h(b)]$$

Example: 4 The value of the integral $\int_{e^{-1}}^{e^2} \left| \frac{\log_e x}{x} \right| dx$ is

[IIT 2000]

(a) 3/2

(b) 5/2

(c) 3

(d) 5

Solution: (b) Put $\log_e x = t \Rightarrow e^t = x$

$$\therefore dx = e^t dt$$

and limits are adjusted as -1 to 2

$$\therefore I = \int_{-1}^2 \left| \frac{t}{e^t} \right| e^t dt = \int_{-1}^2 |t| dt \Rightarrow I = \int_{-1}^0 -t dt + \int_0^2 t dt \Rightarrow I = \left[-\frac{t^2}{2} \right]_{-1}^0 + \left[\frac{t^2}{2} \right]_0^2 \Rightarrow I = 5/2$$

Example: 5 $\int_0^{\pi/2} \frac{dx}{1 + \sin x}$ equals

[MNR 1983; Rajasthan PET 1990; Kurukshetra CEE 1997]

- (a) 0 (b) 1 (c) -1 (d) 2

Solution: (b)

$$I = \int_0^{\pi/2} \frac{dx}{\sin^2 x/2 + \cos^2 x/2 + 2 \sin x/2 \cos x/2}$$

$$I = \int_0^{\pi/2} \frac{dx}{(\sin x/2 + \cos x/2)^2} = \int_0^{\pi/2} \frac{\sec^2 x/2}{(1 + \tan x/2)^2} dx$$

$$\text{Put } (1 + \tan x/2) = t \Rightarrow \frac{1}{2} \sec^2 x/2 dx = dt$$

$$\therefore I = 2 \int_1^2 \frac{dt}{t^2} = -2 \left[\frac{1}{t} \right]_1^2 = -2 \left[\frac{1}{2} - \frac{1}{1} \right] = 1$$

6.4 Properties of Definite Integral

(1) $\int_a^b f(x) dx = \int_a^b f(t) dt$ i.e., The value of a definite integral remains unchanged if its variable is replaced by any other symbol.

Example: 6 $\int_3^6 \frac{1}{x+1} dx$ is equal to

- (a) $[\log(x+1)]_3^6$ (b) $[\log(t+1)]_3^6$ (c) Both (a) and (b) (d) None of these

Solution: (c)

$$I = \int_3^6 \frac{1}{x+1} dx = [\log(x+1)]_3^6, \quad I = \int_3^6 \frac{1}{t+1} dt = [\log(t+1)]_3^6$$

(2) $\int_a^b f(x) dx = -\int_b^a f(x) dx$ i.e., by the interchange in the limits of definite integral, the sign of the integral is changed.

Example: 7 Suppose f is such that $f(-x) = -f(x)$ for every real x and $\int_0^1 f(x) dx = 5$, then $\int_{-1}^0 f(t) dt =$

[MP PET 2000]

- (a) 10 (b) 5 (c) 0 (d) -5

Solution: (d) Given, $\int_0^1 f(x) dx = 5$

$$\text{Put } x = -t \Rightarrow dx = -dt$$

$$\therefore I = -\int_0^{-1} f(-t) dt = -\int_{-1}^0 f(t) dt \Rightarrow I = -5$$

(3) $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, (where $a < c < b$)

or $\int_a^b f(x) dx = \int_a^{c_1} f(x) dx + \int_{c_1}^{c_2} f(x) dx + \dots + \int_{c_n}^b f(x) dx$; (where $a < c_1 < c_2 < \dots < c_n < b$)

Generally this property is used when the integrand has two or more rules in the integration interval.

Important Tips

$$\int_a^b (|x-a| + |x-b|) dx = (b-a)^2$$

6 Definite Integral

Note : $\square$ Property (3) is useful when $f(x)$ is not continuous in $[a, b]$ because we can break up the integral into several integrals at the points of discontinuity so that the function is continuous in the sub-intervals.

$\square$ The expression for $f(x)$ changes at the end points of each of the sub-interval. You might at times be puzzled about the end points as limits of integration. You may not be sure for $x = 0$ as the limit of the first integral or the next one. In fact, it is immaterial, as the area of the line is always zero. Therefore, even if you write $\int_{-1}^0 (1 - 2x) dx$, whereas 0 is not included in its domain it is deemed to be understood that you are approaching $x = 0$ from the left in the first integral and from right in the second integral. Similarly for $x = 1$.

Example: 8 $\int_{-2}^2 |1 - x^2| dx$ is equal to

[IIT 1983; BIT Ranchi 1996; Kurukshetra CEE 1998]

- (a) 2 (b) 4 (c) 6 (d) 8

Solution: (b) $I = \int_{-2}^2 |1 - x^2| dx = \int_{-2}^{-1} |1 - x^2| dx + \int_{-1}^1 |1 - x^2| dx + \int_1^2 |1 - x^2| dx$
 $\Rightarrow I = -\int_{-2}^{-1} (1 - x^2) dx + \int_{-1}^1 (1 - x^2) dx - \int_1^2 (1 - x^2) dx \Rightarrow I = \frac{4}{3} + \frac{4}{3} + \frac{4}{3} = 4.$

Example: 9 $\int_0^{1.5} [x^2] dx$, where $[.]$ denotes the greatest integer function, equals [DCE 2000, 2001; IIT 1988; AMU 1998]

- (a) $2 + \sqrt{2}$ (b) $2 - \sqrt{2}$ (c) $1 + \sqrt{2}$ (d) $\sqrt{2} - 1$

Solution: (b) $I = \int_0^{1.5} [x^2] dx = \int_0^1 [x^2] dx + \int_1^{\sqrt{2}} [x^2] dx + \int_{\sqrt{2}}^{1.5} [x^2] dx \Rightarrow I = 0 + \int_1^{\sqrt{2}} 1 dx + \int_{\sqrt{2}}^{1.5} 2 dx = \sqrt{2} - 1 + 3 - 2\sqrt{2} \Rightarrow I = 2 - \sqrt{2}$

Example: 10 If $f(x) = \begin{cases} e^{\cos x} \cdot \sin x, & |x| \leq 2 \\ 2, & \text{otherwise} \end{cases}$, then $\int_{-2}^3 f(x) dx =$

[IIT 2000]

- (a) 0 (b) 1 (c) 2 (d) 3

Solution: (c) $|x| \leq 2 \Rightarrow -2 \leq x \leq 2$ and $f(x) = e^{\cos x} \sin x$ is an odd function.

$$\therefore I = \int_{-2}^3 f(x) dx = \int_{-2}^2 f(x) dx + \int_2^3 f(x) dx \Rightarrow I = 0 + \int_2^3 2 dx = [2x]_2^3 = 2 \quad [\because \int_{-a}^a f(x) dx = 0 \text{ if } f(x) \text{ is odd and in } (2, 3) f(x) \text{ is } 2]$$

(4) $\int_0^a f(x) dx = \int_0^a f(a - x) dx$: This property can be used only when lower limit is zero. It is generally used for those complicated integrals whose denominators are unchanged when x is replaced by $(a - x)$.

Following integrals can be obtained with the help of above property.

$$(i) \int_0^{\pi/2} \frac{\sin^n x}{\sin^n x + \cos^n x} dx = \int_0^{\pi/2} \frac{\cos^n x}{\cos^n x + \sin^n x} dx = \frac{\pi}{4}$$

$$(ii) \int_0^{\pi/2} \frac{\tan^n x}{1 + \tan^n x} dx = \int_0^{\pi/2} \frac{\cot^n x}{1 + \cot^n x} dx = \frac{\pi}{4}$$

$$(iv) \int_0^{\pi/2} \frac{\sec^n x}{\sec^n x + \operatorname{cosec}^n x} dx = \int_0^{\pi/2} \frac{\operatorname{cosec}^n x}{\operatorname{cosec}^n x + \sec^n x} dx = \frac{\pi}{4}$$

$$(v) \int_0^{\pi/2} f(\sin 2x) \sin x dx = \int_0^{\pi/2} f(\sin 2x) \cos x dx$$

$$(vii) \int_0^{\pi/2} f(\tan x) dx = \int_0^{\pi/2} f(\cot x) dx$$

$$(iii) \int_0^{\pi/2} \frac{1}{1 + \tan^n x} dx = \int_0^{\pi/2} \frac{1}{1 + \cot^n x} dx = \frac{\pi}{4}$$

$$(vi) \int_0^{\pi/2} f(\sin x) dx = \int_0^{\pi/2} f(\cos x) dx$$

$$(viii) \int_0^1 f(\log x) dx = \int_0^1 f[\log(1 - x)] dx$$

$$(ix) \int_0^{\pi/2} \log \tan x dx = \int_0^{\pi/2} \log \cot x dx$$

$$(x) \int_0^{\pi/4} \log(1 + \tan x) dx = \frac{\pi}{8} \log 2$$

$$(xi) \int_0^{\pi/2} \log \sin x dx = \int_0^{\pi/2} \log \cos x dx = -\frac{\pi}{2} \log 2 = \frac{\pi}{2} \log \frac{1}{2}$$

$$(xii) \int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx = \int_0^{\pi/2} \frac{a \sec x + b \operatorname{cosec} x}{\sec x + \operatorname{cosec} x} dx = \int_0^{\pi/2} \frac{a \tan x + b \cot x}{\tan x + \cot x} dx = \frac{\pi}{4} (a + b)$$

Example: 11 $\int_0^{\pi} e^{\sin^2 x} \cos^3 x dx =$

OR

For any integer n , $\int_0^{\pi} e^{\sin^2 x} \cos^3 (2n+1)x dx =$

[MP PET 2002; MNR 1992, 98]

(a) -1

(b) 0

(c) 1

(d) None of these

Solution: (b) Let, $f_1(x) = \cos^3 x = -f(\pi - x)$

and $f_2(x) = \cos^3 (2n+1)x = -f(\pi - x)$

$\therefore I = 0$.

Example: 12 $\int_0^{2a} \frac{f(x)}{f(x) + f(2a-x)} dx$ is equal to

[IIT 1988; Kurukshetra CEE 1999; Karnataka CET 2000]

(a) a

(b) $a/2$

(c) $2a$

(d) 0

Solution: (a) $I = \int_0^{2a} \frac{f(x)}{f(x) + f(2a-x)} dx = \int_0^{2a} \frac{f(2a-x)}{f(2a-x) + f(x)} dx$

$$2I = \int_0^{2a} \frac{f(x) + f(2a-x)}{f(x) + f(2a-x)} dx = \int_0^{2a} 1 dx = [x]_0^{2a} = 2a$$

$\therefore I = a$.

Example: 13 $\int_0^{\pi/2} \frac{\sqrt{\tan x}}{1 + \sqrt{\tan x}} dx = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ is equal to

[MNR 1989, 90; Rajasthan PET 1991, 95]

(a) $\pi/4$

(b) ∞

(c) -1

(d) 1

Solution: (a) We know, $\int_0^{\pi/2} \frac{\tan^n x dx}{1 + \tan^n x} = \frac{\pi}{4}$ for any value of n

$\therefore I = \pi/4$.

$$(5) \int_{-a}^a f(x) dx = \int_0^a f(x) + f(-x) dx.$$

In special case : $\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is even function or } f(-x) = f(x) \\ 0, & \text{if } f(x) \text{ is odd function or } f(-x) = -f(x) \end{cases}$

This property is generally used when integrand is either even or odd function of x .

Example: 14 The integral $\int_{-1/2}^{1/2} \left([x] + \ln \left(\frac{1+x}{1-x} \right) \right) dx$ equal to

(a) $-\frac{1}{2}$

(b) 0

(c) 1

(d) $2 \ln \left(\frac{1}{2} \right)$

Solution: (a) $\log \left(\frac{1+x}{1-x} \right)$ is an odd function of x as $f(-x) = -f(x)$

$$I = \int_{-1/2}^{1/2} [x] dx + 0 \Rightarrow I = \int_{-1/2}^0 [x] dx + \int_0^{1/2} [x] dx \Rightarrow I = \int_{-1/2}^0 -1 dx + 0 \Rightarrow -[x]_{-1/2}^0 = -\frac{1}{2}.$$

8 Definite Integral

Example: 15 The value of the integral $\int_{-1}^1 \log [x + \sqrt{x^2 + 1}] dx$ is [MP PET 2001]

- (a) 0 (b) $\log 2$ (c) $\log 1/2$ (d) None of these

Solution: (a) $f(x) = \log [x + \sqrt{x^2 + 1}]$ is an odd function i.e. $f(-x) = -f(x) \Rightarrow f(x) + f(-x) = 0 \Rightarrow I = 0$.

Example: 16 The value of $\int_{-\pi}^{\pi} (1 - x^2) \sin x \cos^2 x dx$ is [Roorkee 1992; MNR 1998]

- (a) 0 (b) 1 (c) 2 (d) 3

Solution: (a) Let, $f_1(x) = (1 - x^2)$, $f_2(x) = \sin x$ and $f_3(x) = \cos^2 x$

Now, $f_1(x) = f_1(-x)$, $f_2(x) = -f_2(-x)$ and $f_3(x) = f_3(-x)$

$$\therefore I = \int_{-\pi}^{\pi} f(x) dx = \int_{-\pi}^{\pi} [f_1(x) \cdot f_2(x) \cdot f_3(x)] dx = - \int_{-\pi}^{\pi} [f_1(-x) \cdot f_2(-x) \cdot f_3(-x)] dx$$

$$\therefore I = 0$$

$$(6) \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$$

$$\text{In particular, } \int_0^{2a} f(x) dx = \begin{cases} 0, & \text{if } f(2a - x) = -f(x) \\ 2 \int_0^a f(x) dx, & \text{if } f(2a - x) = f(x) \end{cases}$$

It is generally used to make half the upper limit.

Example: 17 If n is any integer, then $\int_0^{\pi} e^{\cos^2 x} \cos^3(2n+1)x dx$ is equal to [IIT 1985; Rajasthan PET 1995]

- (a) x (b) 1 (c) 0 (d) None of these

Solution: (c) $I = \int_0^{\pi} e^{\cos^2(\pi-x)} \cdot \cos^3(2n+1)(\pi-x) dx$

$$\Rightarrow I = - \int_0^{\pi} e^{\cos^2 x} \cdot \cos^3(2n+1)x dx \Rightarrow I = -I$$

$$\Rightarrow 2I = 0 \Rightarrow I = 0.$$

Example: 18 If $I_1 = \int_0^{3\pi} f(\cos^2 x) dx$ and $I_2 = \int_0^{\pi} f(\cos^2 x) dx$ then

- (a) $I_1 = I_2$ (b) $I_1 = 2I_2$ (c) $I_1 = 3I_2$ (d) $I_1 = 4I_2$

Solution: (c) $f(\cos^2 x) = f(\cos^2(3\pi - x))$

$$\therefore I_1 = 3 \int_0^{\pi} f(\cos^2 x) dx \Rightarrow I_1 = 3I_2$$

$$(7) \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

Note: $\square \int_a^b \frac{f(x) dx}{f(x) + f(a+b-x)} = \frac{1}{2}(b-a)$

Example: 19 $\int_{\pi/4}^{3\pi/4} \frac{dx}{1 + \cos x}$ is equal to [IIT 1999]

- (a) 2 (b) -2 (c) 1/2 (d) -1/2

Solution: (a) $I = \int_{\pi/4}^{3\pi/4} \frac{1}{1 + \cos x} dx$ $\left[\because \cos\left(\frac{\pi}{4} + \frac{3\pi}{4} - x\right) = -\cos x \right]$

$$\therefore 2I = \int_{\pi/4}^{3\pi/4} \frac{2}{1 - \cos^2 x} dx$$

$$\Rightarrow 2I = 2 \int_{\pi/4}^{3\pi/4} \operatorname{cosec}^2 x dx \Rightarrow 2I = -2[\cot x]_{\pi/4}^{3\pi/4} = 4 \Rightarrow I = 2.$$

- Example: 20** The value of $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x} + \sqrt{x}} dx$ is [IIT 1994; Kurukshetra CEE 1998]
- (a) 1 (b) 0 (c) -1 (d) 1/2

Solution: (d) $I = \int_2^3 \frac{\sqrt{x}}{\sqrt{5-x} + \sqrt{x}} dx$
 Put $x = 2 + 3 - t \Rightarrow dx = -dt$
 $\therefore I = \int_3^2 \frac{\sqrt{5-t}}{\sqrt{5-t} + \sqrt{t}} (-dt) = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{x}} dx$ and $2I = \int_2^3 \frac{\sqrt{x} + \sqrt{5-x}}{\sqrt{5-x} + \sqrt{x}} dx = \int_2^3 1 dx$
 $\Rightarrow 2I = [x]_2^3 = 1 \Rightarrow I = 1/2$

- Example: 21** If $f(a+b-x) = f(x)$ then $\int_a^b x f(x) dx$ is equal to [Kurukshetra CEE 1993; AIEEE 2003]
- (a) $\frac{a+b}{2} \int_a^b f(b-x) dx$ (b) $\frac{a+b}{2} \int_a^b f(x) dx$ (c) $\frac{b-a}{2} \int_a^b f(x) dx$ (d) None of these

Solution: (b) $I = \int_a^b x f(x) dx$ and $I = \int_a^b (a+b-x) f(a+b-x) dx$
 $\Rightarrow I = \int_a^b (a+b-x) f(x) dx \Rightarrow I = \int_a^b (a+b) f(x) dx - \int_a^b x f(x) dx \Rightarrow 2I = \left[\int_a^b f(x) dx \right] (a+b) \Rightarrow I = \frac{a+b}{2} \int_a^b f(x) dx$

$$(8) \int_0^a x f(x) dx = \frac{1}{2} a \int_0^a f(x) dx \text{ if } f(a-x) = f(x)$$

- Example: 22** If $\int_0^\pi x f(\sin x) dx = k \int_0^\pi f(\sin x) dx$, then the value of k will be [IIT 1982]
- (a) π (b) $\pi/2$ (c) $\pi/4$ (d) 1

Solution: (b) Given, $\int_0^\pi x f(\sin x) dx = k \int_0^\pi f(\sin x) dx$
 $\Rightarrow \int_0^\pi (\pi-x) f(\sin(\pi-x)) dx = k \int_0^\pi f(\sin(\pi-x)) dx \Rightarrow \pi \int_0^\pi f(\sin x) dx - \int_0^\pi x f(\sin x) dx = k \int_0^\pi f(\sin x) dx$
 $\Rightarrow \pi \int_0^\pi f(\sin x) dx - 2k \int_0^\pi f(\sin x) dx = 0 \Rightarrow (\pi - 2k) \int_0^\pi f(\sin x) dx = 0$
 $\therefore \pi - 2k = 0 \Rightarrow k = \pi/2$

(9) If $f(x)$ is a periodic function with period T , then $\int_0^{nT} f(x) dx = n \int_0^T f(x) dx$,

Deduction : If $f(x)$ is a periodic function with period T and $a \in \mathbb{R}^+$, then $\int_{nT}^{a+nT} f(x) dx = \int_0^a f(x) dx$

(10) (i) If $f(x)$ is a periodic function with period T , then

$$\int_a^{a+nT} f(x) dx = n \int_0^T f(x) dx \quad \text{where } n \in \mathbb{I}$$

(a) In particular, if $a = 0$

$$\int_0^{nT} f(x) dx = n \int_0^T f(x) dx, \quad \text{where } n \in \mathbb{I}$$

(b) If $n = 1$, $\int_0^{a+T} f(x) dx = \int_0^T f(x) dx$,

(i) $\int_{mT}^{nT} f(x) dx = (n-m) \int_0^T f(x) dx$, where $n, m \in \mathbb{I}$

(ii) $\int_{a+nT}^{b+nT} f(x) dx = \int_a^b f(x) dx$, where $n \in \mathbb{I}$